ChronoScope: Forecasting Implementations, Contiguous-History Evaluation and Context-Dependent Combination

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Abstract

Forecast comparisons are useful only when their information sets, implementation identities and scoring targets are explicit. ChronoScope combines a diagnostic atlas with a reproducible forecasting workbench. Its fresh final-block experiment executes twenty-two variants on fifteen selected series with recorded package and checkpoint identities. A separate archived-path comparison retains fourteen released model paths alongside five corrected lightweight methods and three matching-expert combinations. Two algorithm defects were corrected: additive HoltWinters seasonal continuation after an incomplete cycle, and a finite-history term in Theta. A separate evaluation defect was identified and invalidated: display-decimated replay histories had been used as consecutive fitting observations. Reconstructing the original observations changes the historical scales and the recomputed predictions. Under the corrected protocol, ContextMemory has mean MASE 1.4504, compared with 1.4489 for RecencyMean and 1.8970 for EqualMean. The additional context component therefore fails its declared aggregate criterion. A source-bound artifact, normative schemas, generated browser types and eight executed bilingual lessons make the implementation and its limits inspectable. One origin per selected series, bounded training configurations and unverified pretraining overlap preclude population superiority or deployment claims.

1 Decision problem and scope

Inventory, staffing and capacity commitments depend on forecasts made before demand is observed. A forecast revision can alter those commitments even when its statistical benefit has not been measured. The relevant engineering question is therefore broader than selecting the smallest displayed error: which observations were available, what algorithm produced the path, which future outcomes were scored, and what uncertainty remains before an action is taken?

ChronoScope supports inspection of these questions through selected real excerpts, generated mechanisms and explicit controls. Generated cases explain mechanisms and expose defects; they are not customer observations. The real excerpts represent electricity demand, air quality and M4 series. Air quality is a forecasting example rather than a demand-planning measurement. No monetary benefit, service-level improvement, laboratory result or customer deployment is inferred from these examples.

The browser offers series structure, forecasts, interval boundaries, individual errors and historical streaming diagnostics. Documentation separates a sourced executive problem framing from

technical method descriptions. The product is a research and teaching instrument. Acceptance of a production policy would additionally require an operational objective, representative data, governance of revisions, latency constraints and prospective evaluation.

2 Implementation identities

The method roster distinguishes algorithms from their implementations. A reference library, a direct adaptation and a small browser-exported model can share an architectural idea while differing in objectives, preprocessing and training budget. Their results remain separate.

Methods	Implementation and settings	Evidence in the archived comparison
SeasonalNaive, SES, Holt, HoltWinters, Theta	NumPy core; browser TypeScript parity; finite smoothing grids	Recomputed on original contiguous prefixes. Gaussian residual intervals are approximations.
AutoETS, AutoTheta, AutoARIMA	statsforecast; ARIMA stepwise and approximate, context cap 360	Released paths rescored. Separate worked lessons freshly execute the installed adapters.
LightGBM	MLForecast lag table; 100 trees, 31 leaves, learning rate 0.05, context cap 500	Released point-regression path rescored; Gaussian residual intervals. A lesson freshly executes the adapter.
NLinear, DLinear, NHITS	Direct-torch quantile-head adaptations	Released paths rescored; not asserted to be exact replications of the papers.
NLinear (nf), DLinear (nf), NHITS (nf)	neuralforecast reference implementations; fixed 300-step training budget	Released paths rescored; the budget is part of the experiment.
Chronos-Bolt, Chronos-2, TimesFM 2.5, TiRex-2	Pretrained checkpoint adapters; TiRex bridge uses WSL/CUDA	Released paths rescored, without current-checkpoint or pretraining-independence claims.
ContextMemory, RecencyMean, EqualMean	Same five light experts; fixed causal memory and two ablations	Newly computed with matching original prefixes and final targets.

An additional browser NLinear model is exported to ONNX after least-squares fitting on generated windows. It predicts twelve observations from a forty-eight-observation context. It is an independently identified live demonstration, not an additional member of the twenty-two-variant aggregate experiment. On real cases its first twelve held-out targets can be inspected without adapting the fit to those targets. Its interval scale is inherited from its training metadata and does not establish calibration on the real excerpts.

2.1 Corrected lightweight methods

SeasonalNaive repeats the latest observed cycle. SES updates a level, Holt adds a trend, and additive HoltWinters additionally updates a seasonal component. At a prefix length n, the seasonal slot for zero-based future lead k must continue as $(n + k) \bmod m$. Restarting at $k \bmod m$ is incorrect when n is not a multiple of m. An adversarial partial-season example tests this condition independently in Python; the browser parity fixture includes a non-multiple history and a horizon longer than one cycle.

For the nonseasonal Theta implementation, the corrected point equation is

$$\widehat{y}_{n+h} = \ell_n + \frac{b}{2}\left[h - 1 + \frac{1-(1-\alpha)^n}{\alpha}\right], \qquad h \geq 1, \tag{1}$$

where ℓ_n is the selected SES level and b is the fitted linear slope. An independent two-Theta-line construction verifies the finite-history term following Hyndman and Billah [1]. This local method does not inherit AutoTheta's automatic seasonal adjustment merely because the names are related.

The Gaussian widening rule maps a residual standard deviation to three ordered columns at probabilities 0.1, 0.5 and 0.9. Residual estimation, Gaussian shape and horizon widening are assumptions. Neither an ordered matrix nor a quantile-loss fit establishes marginal or conditional calibration. These distinctions also apply to the direct neural adaptations and the lag-tree adapter.

3 Diagnostic atlas and computational boundaries

The diagnostic toolkit contains ten families: stationarity, autocorrelation, seasonality, filtering and decomposition, change points, volatility, distribution and complexity, fractals, nonlinear dynamics, and causality. Methods include ADF/KPSS and related tests, ACF/PACF and Ljung–Box diagnostics, STL/MSTL and periodograms, offline change-point procedures, ARCH/GARCH diagnostics and surrogate-based nonlinear analyses. A short series or unavailable dependency produces an explicit unavailable result for that analysis; it does not silently produce a substitute measurement.

Statistical diagnostics are descriptive or model-dependent. An autocorrelation band containing the observed coefficients does not prove white noise. A unit-root test does not certify future stationarity. A fractal estimate is sensitive to sample length, detrending and the assumed stochastic class. A nonlinear surrogate rejection does not uniquely identify deterministic chaos. The workbench exposes these cautions alongside the results rather than presenting a diagnostic label as a forecasting guarantee.

The browser recomputes lightweight diagnostics and light forecasts for generated scenarios. Committed analysis artifacts retain their original scope; several describe the complete observed series, including what later becomes a forecasting holdout. They are suitable for retrospective explanation, not for feature selection at the earlier forecast origin. The proposed combination computes its own prefix-only contexts and does not consume those retrospective diagnostics.

Heavy statistical, neural and pretrained calculations run offline. GPU availability, checkpoint caches and Python environments are distinct from the served browser bundle. A local environment with only NumPy cannot be represented as a complete neural or foundation-model rerun. Conversely, rescoring a stored model path does not require pretending that its original runtime has been restored.

4 Data provenance and temporal reconstruction

The corpus contains eleven generated cases and four real excerpts. Generated mechanisms cover seasonality, trend, intermittent arrivals, a level shift, multiple seasonalities, heteroscedasticity, long memory, a delayed nonlinear system and a known-future exogenous driver. White noise and a near-random walk remain visible as controls. The exogenous-driver comparison requires a declared arrival policy; univariate forecasts do not gain access to future covariates automatically.

Table 2: Admitted cases: original contiguous history length, horizon and seasonal lag.

Case	History	Horizon	Lag	Source
BRKV_level_shift	288	12	12	Generated
CHAO_mackey	488	12	1	Generated
CTRL_white_noise	148	12	1	Generated
EXOG_promo	226	14	7	Generated
HETV_garch	488	12	1	Generated
INTM_demand	188	12	1	Generated
LMEM_fractional	488	12	1	Generated
MSEA_daily_weekly	648	24	24	Generated
REAL_electricity	456	24	24	Real excerpt
REAL_m4_daily	386	14	7	Real excerpt
REAL_m4_hourly	456	24	24	Real excerpt
REAL_pm25	456	24	24	Real excerpt
RWLK_noise	188	12	1	Generated
SEAS_hourly	456	24	24	Generated
TRND_seasonal	168	12	12	Generated

The source manifest records provenance, redistribution permission, seed, original series descriptors, artifact paths and byte counts. Real-data CSVs are committed as compact admitted excerpts. A source's redistribution status is an explicit field; a restricted source must not expose its raw path merely because aggregate metrics are publishable. The current comparison uses only available, non-redacted inputs.

4.1 Why a replay history is not a fitting series

The released trace stores at most three hundred historical display samples and their original indices. Nine case histories exceed that display limit. Treating those samples as consecutive observations changes the time interval and can destroy the meaning of a seasonal lag. It also changes a historical scale used by MASE. The initial comparison made this error and is invalidated; its earlier bytes remain identifiable through Git, not as admitted scientific evidence.

The corrected protocol reconstructs the original series from each committed CSV or deterministic case definition and its manifest seed. At every displayed index the reconstructed observation must agree with the released rounded value within 0.000051. The final targets must also agree at that tolerance, and the original prefix length must match the trace's declared history length. All fifteen reconstructions satisfy these checks. The complete contiguous prefix is used for light fitting and historical scaling. Original per-engine context caps remain explicit; a common available prefix does not mean every algorithm consumes the same number of observations.

The reconstructed history retains source precision. Frozen targets and model paths retain their released four-decimal precision. This small rounding distinction is disclosed and is not the much larger temporal-decimation error. Source and implementation hashes bind the corrected artifact to the exact inputs. A replay-only array whose declared length exceeds its display length is refused as a consecutive fit input.

5 Common final-block protocol

Each selected series supplies one forecast origin and one final block. Fourteen heavy forecast paths remain frozen; five corrected lightweight methods and three combinations are newly computed. Every admitted path is scored on the same actual values and horizon within its case. Original rolling-backtest results remain archived. Their differing origin budgets, ranging from three to more than twenty-five windows, make a direct ranking of those aggregates confounded by the evaluation schedule.

Let $y_1, \ldots, y_n$ denote the contiguous prefix, $a_1, \ldots, a_H$ the final targets and p_j a point prediction. The historical seasonal scale is

$$s_m = \frac{1}{n-m} \sum_{t=m+1}^{n} |y_t - y_{t-m}|, \tag{2}$$

with the lag bounded to the available history. MAE is $H^{-1} \sum_j |a_j - p_j|$, RMSE is the square root of mean squared error, and MASE is MAE/s_m [2]. A value below one compares a future error with a historical scale. It does not directly show that the method beat an independently scored SeasonalNaive path on the same future block. A zero scale produces an unavailable scaled metric rather than an invented denominator.

For lower and upper columns l_j, u_j, empirical coverage is $H^{-1} \sum_j \mathbf{1}\{l_j \le a_j \le u_j\}$ and width is $H^{-1} \sum_j (u_j - l_j)$. With nominal interval miscoverage $\alpha = 0.2$, the interval score is

$$IS_j = (u_j - l_j) + \frac{2}{\alpha}(l_j - a_j)_+ + \frac{2}{\alpha}(a_j - u_j)_+. \tag{3}$$

MSIS is its block mean divided by s_m. The artifact's WQL uses only $Q = \{0.1, 0.5, 0.9\}$:

$$\mathrm{WQL}_3 = \frac{2 \sum_{q \in Q} \sum_{j=1}^{H} \rho_q(a_j - \widehat{y}_j^{(q)})}{3 \sum_{j=1}^{H} |a_j|}, \quad \rho_q(e) = \max(qe, (q-1)e). \tag{4}$$

This is an average of three quantile losses, not a complete predictive-density evaluation. A zero absolute-target sum makes this normalized score unavailable. Proper-score theory motivates reporting probabilistic quality, while the finite quantile grid limits what is observed [3].

6 A fixed context-memory proposal

ContextMemory tests whether similar completed historical contexts help estimate expert competence beyond recency alone. Feature-dependent combinations and dynamic ensembles are established ideas [4, 5]. The experiment tests a specific within-series recurrence rule and makes no priority claim for contextual weighting.

The expert set is fixed to SeasonalNaive, SES, Holt, HoltWinters and Theta. Up to twelve completed historical origins are selected within the prefix. For every origin o, eligibility requires $o + H \le n$. A normalized slope, level shift, volatility-change statistic and seasonal correlation describe the context. Expert multi-step absolute losses are normalized using information available at that origin. A Gaussian context-distance kernel combines with recency weights of half-life six origins. Local estimated losses are shrunk 25% toward the recency-only loss estimate.

An exponential loss transformation produces a simplex allocation, followed by a 5% uniform floor. With $K = 5$ experts, each final weight is at least $0.05/K = 0.01$ and the weights sum to one. Corresponding point and quantile columns are averaged with these same weights. Nonnegative convex weights preserve the order of the input quantile columns. They do not make the result a calibrated predictive mixture.

RecencyMean removes context similarity while retaining the experts, historical origins, recency weighting and competence transformation. EqualMean assigns a uniform allocation. These matching ablations distinguish the contextual component from the more general benefit of combining methods. A positive scale and location transformation supplies a structural check away from numerical degeneracy thresholds. Completed-target eligibility and quantile ordering have independent adversarial tests.

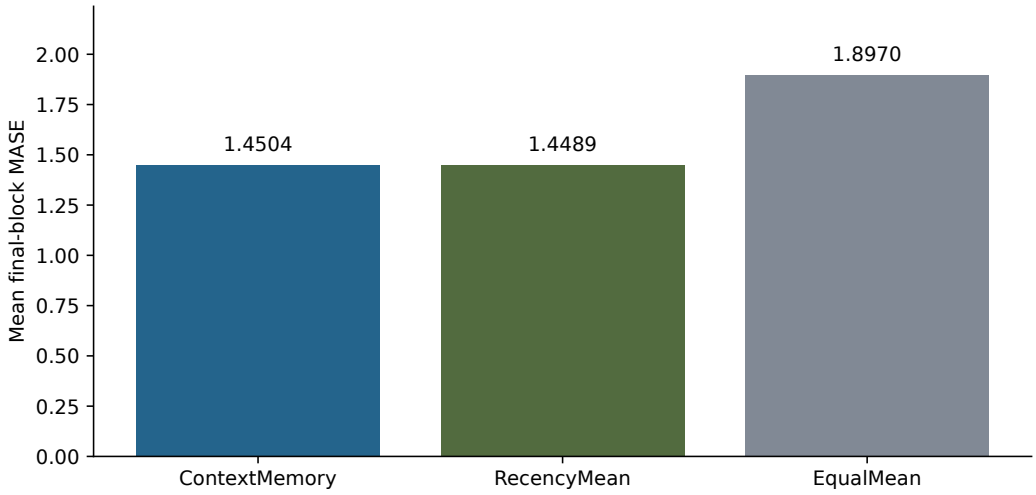


Figure 1: Fixed matching-expert combinations on the corrected fifteen-case protocol. Lower mean MASE is preferable. The contextual component does not improve on the recency-only ablation. Bars summarize selected final blocks, not independent population replications.

7 Archived-path final-block results

Table 3: Corrected common final-block comparison. Means weight each selected case equally.

Method	Cases	Mean MASE	Mean coverage
Chronos-Bolt	15	1.0363	0.7563
TimesFM-2.5	15	1.0633	0.8071
TiRex-2	15	1.0809	0.7694
Chronos-2	15	1.1555	0.6905
DLinear	15	1.1600	0.5595
AutoARIMA	15	1.2104	0.7813
NLinear	15	1.2298	0.4524
NHITS (nf)	15	1.2546	0.5079
NHITS	15	1.2898	0.4857
LightGBM	15	1.2983	0.5786
SeasonalNaive	15	1.4137	0.8702
RecencyMean	15	1.4489	0.8889
ContextMemory	15	1.4504	0.8944
AutoTheta	15	1.4964	0.7329
HoltWinters	15	1.6778	0.8639
AutoETS	15	1.6940	0.7639
NLinear (nf)	15	1.7547	0.1821

Table 3: Corrected common final-block comparison. Means weight each selected case equally.

Method	Cases	Mean MASE	Mean coverage
EqualMean	15	1.8970	0.8611
SES	15	1.9167	0.8274
Theta	15	1.9224	0.8274
Holt	15	3.9846	0.7302
DLinear (nf)	15	10.7385	0.4996

Chronos-Bolt has the smallest mean MASE in this selected, mixed-provenance comparison, followed by TimesFM 2.5 and TiRex-2. Their paths were not freshly refitted or regenerated for this experiment. No statistical significance, resource-normalized superiority or independence from pretraining data follows from that order. Method-specific training budgets and context limits remain part of the compared configurations.

Coverage exposes substantial differences that a point ranking conceals. Several neural paths have narrow intervals and low observed coverage. A broad interval can increase coverage at the expense of decision usefulness; coverage alone should not be optimized. Individual case outcomes and widths remain available in the artifact and workbench, including unfavorable neural and combination results.

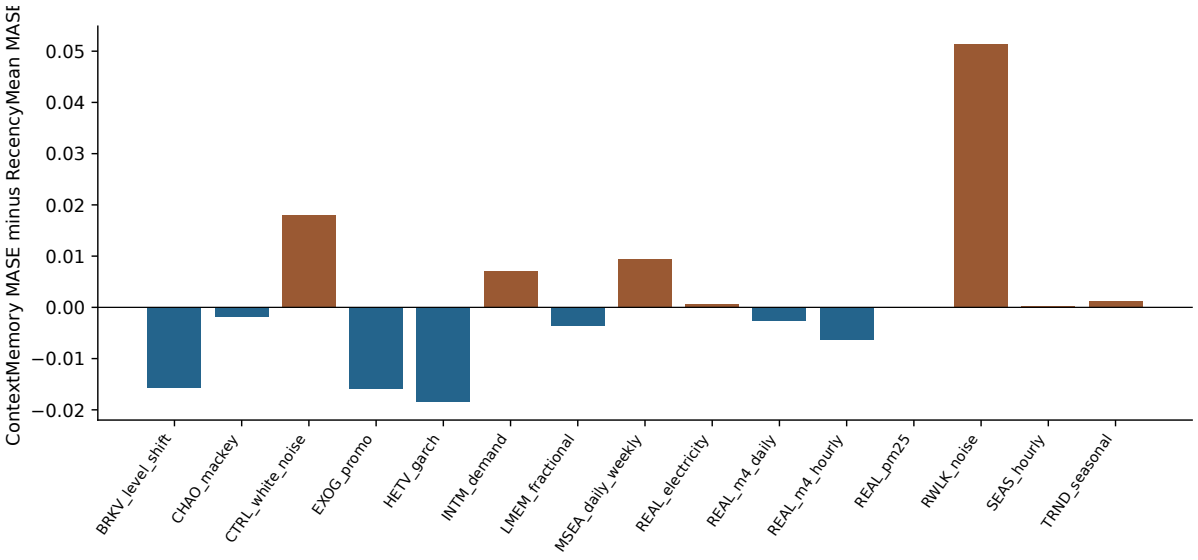


Figure 2: ContextMemory minus RecencyMean MASE for every selected case. Positive values favor the recency-only ablation. The common target and expert set isolate the contextual weighting component within each case, but the selected origins do not establish a general effect.

ContextMemory's mean MASE is 1.4504, versus 1.4489 for RecencyMean. The difference is unfavorable at the declared primary criterion, despite occasional individual improvements. EqualMean gives 1.8970. Settings are unchanged after either the invalidated display-history calculation or the corrected run. A changed configuration would define a new experiment and must not overwrite this result.

8 Fresh execution of the complete roster

A second experiment fits or infers all twenty-two configurations again from the same verified contiguous prefixes. The separately admitted artifact records every prediction, source-file and input hash, checkpoint revision, package version, fit/inference duration and warning category. An independent standard-library checker recomputes all seven metrics from the stored numerical arrays, verifies complete inventories and rejects trials, changed targets, frozen paths disguised as new fits and stale source bindings. These admission checks establish internal numerical consistency and recorded identity; they do not establish independence from model pretraining data.

The execution uses Python 3.12, NumPy 2.1.3, PyTorch 2.9.1 with CUDA 12.8, StatsForecast 2.0.3, LightGBM 4.6.0, MLForecast 1.0.31, NeuralForecast 3.1.9, Chronos forecasting 2.3.2, TimesFM 2.0.2 and TiRex-2 0.1.1. Its GPU is an NVIDIA GeForce RTX 4070 Laptop GPU. NumPy follows the TiRex-compatible line in this isolated Linux environment; notebook/parity computations use their separately pinned environment. Direct and reference neural models retain their declared 300-step budgets. LightGBM uses one worker; BLAS/OpenMP threads and compiler workers are bounded, and torch CPU operations use four threads. These resource choices are part of the recorded configuration.

Table 4: Fresh execution alongside the archived-path comparison. The same fifteen final blocks are used in both; these are not independent replications.

Method	Archived MASE	Fresh MASE	Fresh coverage
Chronos-Bolt	1.0363	1.0363	0.7563
TimesFM-2.5	1.0633	1.0633	0.8071
TiRex-2	1.0809	1.0809	0.7694
Chronos-2	1.1555	1.1555	0.6905
DLinear	1.1600	1.1600	0.5595
AutoARIMA	1.2104	1.2104	0.7813
NLinear	1.2298	1.2298	0.4524
LightGBM	1.2983	1.2983	0.5786
NHITS (nf)	1.2546	1.2993	0.4968
NHITS	1.2898	1.3031	0.4746
SeasonalNaive	1.4137	1.4137	0.8702
RecencyMean	1.4489	1.4489	0.8889
ContextMemory	1.4504	1.4504	0.8944
AutoTheta	1.4964	1.4964	0.7329
HoltWinters	1.6778	1.6778	0.8639
AutoETS	1.6940	1.6940	0.7639
NLinear (nf)	1.7547	1.7547	0.1821
EqualMean	1.8970	1.8970	0.8611
SES	1.9167	1.9167	0.8274
Theta	1.9224	1.9224	0.8274
Holt	3.9846	3.9846	0.7302
DLinear (nf)	10.7385	10.7385	0.4996

The two experiments use the same final targets, so rerunning an implementation is not another independent statistical test. Differences can reflect runtime, precision or training implementation rather than a changed data population. ContextMemory still fails its aggregate comparison with RecencyMean; the selected settings are not retuned after that outcome. Warning categories remain

visible even when predictions are finite. Durations include fitting, inference, imports and first-use loading and cannot be interpreted as a normalized warm-inference leaderboard.

Exact Hugging Face revisions identify the four admitted pretrained checkpoints. TimesFM 3.0 is a newer upstream candidate with noncommercial, nonproduction weight terms; the admitted experiment retains the Apache-licensed 2.5 checkpoint. This roster therefore makes no claim to contain the latest released model of every family. The original streaming results below were not regenerated by this final-block execution.

9 Streaming evaluation as separate evidence

The committed streaming experiments use a predict–observe–update order inspired by prequential evaluation [6]. Their raw, adaptive conformal and PID-controlled intervals are evaluated along observation arrivals. Covariate availability is governed by an explicit known-future or lagged-past policy. These trajectories answer a different question from the final-block experiment and must not be pooled with its scores.

Adaptive conformal inference responds to distribution shift, while conformal PID control provides another adaptive mechanism [7, 8]. Their assumptions and guarantees do not translate into conditional coverage for every local regime. Rolling coverage, width and adaptation delay should be read together. Original streaming timings are historical observations of their original runtime, rather than contemporary browser latency or a normalized hardware benchmark.

10 Artifact and browser engineering

Nine catalog boundaries cover traces, manifests, the index, diagnostic analysis, streaming results, ONNX metadata/assets, the archived-path comparison, fresh execution and historical selector metadata. Normative JSON Schemas validate the committed JSON documents and generate browser TypeScript types through an OpenAPI component inventory. That inventory supports type generation without activating a prediction API. Cross-field invariants, source hashes, temporal reconstruction and numerical parity remain executable tests; a structural schema alone cannot enforce them.

The browser delegates chart interaction, summaries, tables, reset controls and keyboard support to the pinned shared shell. The product supplies data adapters, stable method identities and scenario state. Bilingual content uses locale-aware numbers; model and licence versions retain their identity. Request cancellation and generation identifiers prevent a slower case request or inference result from overwriting a newer selection.

Numbered local workflows check prerequisites, initialize declared environments, build an external data candidate and start the development server. Generation binds source and payload fingerprints, refuses partial inventories and changed frozen inputs, and separates candidate creation from explicit promotion. Historical engine-version metadata is retained rather than relabeled to match the new browser release. An unused epidemic-model surrogate inherited from an older repository structure was removed.

Eight executed English and Spanish notebooks progress through a seasonal reference, an incomplete-season experiment, actual statistical/tree adapter execution and the fixed combination with its ablations. Their metadata records exact direct dependencies, sources, inputs and method coverage. Fresh-kernel execution and byte comparison check that committed outputs are current. Numerical convergence notices are retained as portable text; machine paths and incidental notebook metadata are excluded.

11 Validation and remaining scientific limits

The validation layers are distinct. Structural schema checks establish artifact shape; source bindings establish input identity; independent equation tests establish selected numerical behaviours; browser traversal measures rendered routes and interactions; local notebook reproduction measures executable teaching artifacts. A successful build alone establishes none of the other layers. Computational records distinguish local numerical checks from remote CI, deployment and owner acceptance.

The main scientific limitations are: one origin per selected series; a small, non-random corpus; configuration-specific runtime and historical evidence retained in a separate experiment; unverified pretrained-data overlap; bounded historical training budgets; approximate lightweight uncertainty; possible numerical convergence issues in freshly executed reference adapters; and retrospective diagnostic artifacts that may include the final block. No operational acceptance or economic benefit is implied. A broader study should predeclare datasets, repeated origins, runtime and memory budgets, dependence-aware comparisons and independent pretraining-overlap checks.

For the contextual proposal, a stronger follow-up also needs faithful FFORMA and arbitrated-ensemble comparators rather than only internal ablations. Such work could reject or improve the proposed recurrence rule. The current negative outcome remains informative without being promoted to a general impossibility result.

12 Reproduction and availability

The source tree contains the corrected computation in `chronoscopelab.reevaluation`, the fixed proposal in `model/context_memory.py`, original released traces, admitted source CSVs and the generated comparison. Running `scripts/reevaluate.py` regenerates the common final-block artifact while preserving the original traces. The staged local workflow provides complete candidate admission and explicit promotion. Exact commands, environment pins and validation results are recorded with the source.

The numerical tables and figures are generated directly from the two admitted comparison artifacts. Code and compact admitted data use the repository's software/data licences; prose uses the licence stated in the header. External publication, remote promotion and a verified live deployment are separate operations. Absence of those operations does not change a local numerical result, but prevents representing the candidate as a publicly delivered release.

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